

Stakeholder Engagement Plan

AgriSLM - Specific Small Language Model for Agriculture and Climate

Final Submission - Team QueenElizabeth

1. Team and Project Details

Team name	QueenElizabeth
Team members	Ernest Nchabeng and Andrew Devadason
Project name	AgriSLM - Specific Small Language Model for Agriculture and Climate
Stakeholder groups	Smallholder farmers; Agricultural extension officers
Engagement focus	Inclusive, locally relevant and responsible agricultural AI decision support

2. Project and Engagement Purpose

AgriSLM is a domain-specific small language model project for agriculture and climate. This stakeholder engagement plan moves the project from stakeholder analysis to meaningful participation. Following the supplied task, the plan selects two stakeholder groups, identifies what matters to them, considers potential harms and benefits, chooses an engagement method for each group, and explains how stakeholder input can shape the dataset and model. The engagement approach is intended to help the team identify risks early, improve local relevance and usability, and ensure that the AI solution is developed with rather than merely for its intended users.

3. What Matters Most to the Two Stakeholder Groups (approximately 100 words)

Smallholder farmers need practical, trustworthy and locally relevant agricultural guidance that reflects their crops, climate, soil, water conditions, languages and farming practices. They value accessible information that supports better decisions without ignoring local knowledge or creating unnecessary privacy risks.

Agricultural extension officers need scientifically credible, explainable and current information that supports rather than replaces their professional role. They value reliable sources, regional accuracy, clear uncertainty and safeguards around higher-risk advice such as pesticide, fertiliser and disease recommendations. Both groups need AgriSLM to be inclusive, understandable, accountable and responsive to real agricultural and community needs.

4. Stakeholder Group 1 - Smallholder Farmers

Potential harms

- Inaccurate or poorly localized recommendations that could lead to inappropriate farming decisions.
- Exclusion or misrepresentation of local languages, farming practices and community knowledge.
- Privacy or consent problems when farmer or community information is collected.
- Digital exclusion where connectivity, devices or digital literacy are limited.
- Over-reliance on AI when field inspection or professional agricultural advice is necessary.

Potential benefits

- Faster access to useful agricultural and climate information.
- Better-informed decisions about crops, pests, soil, water and climate risks.
- Improved access to knowledge for underserved farming communities.
- Greater representation of farmers' language, experience and priorities in the AI system.
- More confidence and trust when users can see that their feedback influences the solution.

Engagement method - Participatory Design

Participatory Design is appropriate for smallholder farmers because AgriSLM is intended to provide agricultural guidance that must work in real farming contexts, not only in a laboratory or technical demonstration. The supplied task document identifies Participatory Design as an engagement method and gives the example of co-design workshops in which intended users contribute examples and review model outputs. For AgriSLM, the QueenElizabeth team would organize small, accessible co-design sessions with farmers and use practical farming scenarios rather than highly technical AI terminology.

During the sessions, farmers would describe the questions they commonly ask about crop diseases, pests, rainfall, irrigation, soil condition, planting, harvesting and climate risk. They could explain the words and expressions they naturally use, the information they normally trust, and the difficulties they face when seeking agricultural advice. The team could then present sample AgriSLM questions and answers and ask participants to assess whether the outputs are understandable, useful, culturally appropriate and realistic. Farmers could identify confusing terminology, missing local context or recommendations that would be difficult to apply with available resources.

Participation should be intentionally inclusive. Recruitment should seek variation in location, crop type, farm scale, age, gender and digital experience where feasible. This helps reduce the risk that the dataset and interface reflect only the most visible or digitally connected farmers. Participation should be voluntary, with clear explanation of how feedback, examples or local knowledge may be used. Where community or traditional knowledge is contributed, provenance, consent and appropriate reuse conditions should be recorded rather than treating that knowledge as automatically available for model training.

The engagement should produce concrete design outputs. These could include a prioritized list of farmer questions, examples of preferred answer length and language, common local terms, accessibility needs, important safety concerns and cases in which the model should recommend professional support. The team can convert these findings into dataset requirements, evaluation prompts and usability criteria. After revisions, selected participants can review updated outputs to determine whether their earlier feedback was meaningfully incorporated.

This method fits AgriSLM because technical accuracy alone does not guarantee usefulness. A response may be scientifically plausible yet fail because it is unclear, inaccessible or disconnected from local conditions. Participatory Design makes farmers contributors to the development process rather than passive recipients of technology. It can improve relevance, representation, trust and usability while exposing harms before wider deployment. The method therefore supports responsible innovation by connecting model design to the lived realities of the people the system is intended to benefit.

5. Stakeholder Group 2 - Agricultural Extension Officers

Potential harms

- Hallucinated, outdated or scientifically weak agricultural guidance.
- Recommendations that are not suitable for the relevant region, crop or agro-ecological context.
- Unsafe advice involving pesticides, fertilisers, disease diagnosis or other higher-risk decisions.
- Loss of professional trust if AI outputs contradict reliable extension guidance without evidence.
- Farmers treating general AI information as a substitute for professional diagnosis or field assessment.

Potential benefits

- Faster access to relevant agricultural information for routine farmer questions.
- Improved farmer outreach and an additional decision-support resource.
- Reduced repetition of common information requests.
- A structured mechanism for identifying knowledge, safety and dataset gaps.
- Better alignment between AI outputs and professional extension practice.

Engagement method - User Committee

A User Committee is appropriate for agricultural extension officers because AgriSLM needs continuing professional review, not only one-off consultation. The supplied task document lists User Committees among the engagement methods that can be selected according to project needs. For QueenElizabeth, a small advisory committee could include extension officers and, where appropriate, other agricultural specialists familiar with the intended farming contexts. Its role would be to review representative AgriSLM outputs and advise the team on accuracy, relevance, safety and professional usability.

The committee would meet at defined development milestones and review a structured sample of model responses covering topics such as crop disease, pests, soil health, water management, fertiliser use and climate adaptation. Members could assess whether an answer is factually sound, sufficiently localized, understandable to farmers, supported by suitable evidence and appropriately cautious when information is incomplete. They could also identify situations where AgriSLM should ask for more context or direct the user to a qualified professional rather than giving a confident recommendation.

Higher-risk outputs would receive particular attention. For example, a committee member might flag a pesticide recommendation that lacks a confirmed diagnosis, a fertiliser rate that ignores soil testing, or a disease response that presents uncertainty as certainty. Such findings can become explicit safety rules and evaluation cases. The committee could also review whether the model's sources are current and appropriate and identify information that should be removed, updated or supplemented.

The User Committee would also help identify representation gaps in the dataset. If AgriSLM performs well for one crop, region or farming condition but poorly for another, extension officers can recommend additional authoritative material or testing scenarios. Their practical experience can help the team distinguish between a general agricultural principle and advice that requires local adaptation. Committee feedback should be documented with the issue identified, recommended action, team response and status of the change. This creates an accountability trail rather than informal feedback that may be forgotten.

Importantly, the committee should strengthen rather than replace the relationship between farmers and extension services. AgriSLM should be positioned as decision support that can make information easier to access while preserving the role of professional judgment and field assessment. Periodic review after model

updates would allow the committee to detect new problems and verify whether previous concerns were resolved. This ongoing structure makes the User Committee a good fit for AgriSLM because it combines professional oversight, dataset quality improvement, safety review and continuous accountability throughout the AI development lifecycle.

6. How the Engagement Methods Will Influence the Dataset and Model
(approximately 100 words)

Farmer participatory-design sessions will reveal real questions, terminology, languages, local practices, accessibility needs and underrepresented farming conditions, allowing QueenElizabeth to improve dataset coverage and model usability. Agricultural extension officers will review scientific accuracy, source quality, regional relevance, uncertainty and potentially unsafe recommendations. Their feedback can identify records that require correction, removal or expansion and can create stronger evaluation and safety cases. Together, the two methods form a feedback loop: community experience improves relevance and representation, while professional agricultural expertise strengthens reliability and safeguards. The resulting AgriSLM should therefore become more inclusive, transparent, useful and accountable.

7. Reflection - How Stakeholder Engagement Will Improve Our AI Solution

Stakeholder engagement changes AgriSLM from a solution designed mainly from technical assumptions into one shaped by the people who will use, interpret and support it. Smallholder farmers can show the QueenElizabeth team whether the system reflects real questions, understandable language, local practices and practical constraints. Their participation can reveal needs or harms that may not be visible in the original dataset. Agricultural extension officers add a different but complementary perspective by testing scientific credibility, identifying unsafe or overconfident advice, checking regional relevance and clarifying when professional assessment is required.

This engagement will improve both the data and the model. Farmer feedback can lead to better representation, more realistic prompts and clearer outputs, while extension review can improve source selection, factual evaluation, safety rules and uncertainty handling. Most importantly, the process creates accountability: feedback is collected, documented, translated into design changes and reviewed again. For our team, responsible AI therefore means more than producing a technically capable model. It means developing AgriSLM through continuous dialogue so that usefulness, fairness, safety and local relevance are evaluated by the communities and professionals whose lives and work may be affected by the system.

8. Engagement Action Summary

Stakeholder	Priority	Key risk	Expected benefit	Method	How feedback is used
Smallholder farmers	Local, practical and accessible guidance	Exclusion or inappropriate advice	Better-informed farming decisions	Participatory Design	Questions, language, usability and representation requirements
Agricultural extension officers	Accuracy, evidence, safety and professional	Unsafe or misleading guidance	Stronger decision support and outreach	User Committee	Source review, safety cases, dataset gaps and model

Stakeholder	Priority	Key risk	Expected benefit	Method	How feedback is used
	relevance				evaluation

9. Submission Readiness Checklist

Requirement	Status
Team name included	Yes - QueenElizabeth
Team members included	Yes - Ernest Nchabeng and Andrew Devadason
Project details included	Yes - AgriSLM
Two stakeholder groups selected	Yes
Stakeholder priorities described	Yes
Potential harms and benefits identified	Yes
One engagement method selected for each group	Yes - Participatory Design and User Committee
Rationale for each method provided	Yes
Dataset and model influence described	Yes
Explicit reflection on improving the AI solution	Yes

10. Source Basis

This submission is based on the supplied task document, “Design a mini-engagement plan.” The task requires selection of two stakeholder groups; a description of what matters most to them; identification of potential harms and benefits; selection of engagement methods such as Participatory Design or User Committees; explanation of why each method fits; and a description of how engagement can influence the dataset and model design. This final version also explicitly includes the requested team/project details and reflection.